

GARD: Geometric Abstention via Robust Disagreement for Safe Large Language Models

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Abstract

Hallucination detection in large language models (LLMs) faces a fundamental tension: permutation based uncertainty estimation methods optimize expected correctness under the model distribution but fail catastrophically on adversarial inputs, while existing geometric approaches conflate evidence disagreement with insufficient support. We introduce **GARD** (*Geometric Abstention via Robust Disagreement*), a framework grounded in Geometric Algebra that *separates* conflict detection from weakness detection via orthogonal geometric observables. We prove that sign instability of invariant alignment margins defines a provably unsafe exclusion region, independent of application preferences. Evaluating on an adversarial benchmark designed to expose permutation method failures, we demonstrate that: (i) permutation dispersion is anti-correlated with adversarial risk (below random performance) and provides no mechanism for safety-coverage tradeoff control; (ii) GARD exposes a *controllable Pareto frontier* where users explicitly trade conflict detection (ranging from near-complete at safety-first to moderate at coverage-first) against paraphrase acceptance; (iii) this tradeoff arises from empirical overlap in embedding space coherence, not theoretical insufficiency. As a risk-controlled abstention system rather than a ranking classifier, GARD provides deterministic linear-time computation with worst-case robustness guarantees and policy-adjustable safety constraints, offering a principled path toward auditable deployment in safety-critical applications.

1 Introduction

1.1 Motivation

Retrieval-augmented generation (RAG) systems process multiple evidence chunks to answer user queries, yet suffer from *hallucination*—confidently incorrect outputs unsupported by the evidence [9, 10]. This phenomenon is exacerbated by LLMs’ sensitivity to evidence order: permuting semantically exchangeable inputs can flip predictions [5, 11], violating the permutation invariance expected of set-based reasoning.

Current hallucination detection methods fall into two paradigms. *Sampling-based* approaches estimate uncertainty via permutation averaging [4], semantic entropy over generations [4], or Monte Carlo dropout [6]. These methods optimize *average-case correctness* under the model distribution $\mathbb{E}_{\text{model}}[\text{correct}]$, offering no guarantees against adversarial or worst-case failures. *Information-theoretic* analyses characterize hallucination as inevitable under finite capacity [1, 2], but provide no constructive detection mechanism.

The Central Problem. Existing methods conflate two distinct failure modes:

1. **Evidence conflict:** Internal disagreement between chunks (e.g., “Einstein born 1879” vs. “Einstein born 1880”).
2. **Insufficient support:** Weak alignment between evidence and query, leaving the answer vulnerable to small perturbations.

Standard scalar margin-based approaches treat these identically, leading to systematic failures: accepting confident but conflicted answers (false negatives on conflicts) or rejecting diverse but consistent paraphrases (false positives).

1.2 Our Contribution

We develop a *geometric* theory of hallucination that:

1. **Separates conflict from weakness** via orthogonal observables in Geometric Algebra: *bivector coherence* $\mathcal{C}$ measures geometric disagreement, while *GISR* (Geometric Insufficiency-Sufficiency Ratio) measures margin robustness.
2. **Proves necessity of abstention** via worst-case analysis: $\text{GISR} < 1$ defines a provably unsafe exclusion region where adversarial perturbations can flip the answer sign.
3. **Exposes controllable safety-coverage frontier:** On CLOT Bench (200 adversarial examples), GARD traces a Pareto curve from safety-first (96.4% conflict recall, 0% paraphrase acceptance) to coverage-first (40% conflict recall, 100% paraphrase acceptance). Users select operating points via threshold C_{hi} based on application risk tolerance.
4. **Validates failure of permutation methods:** PermProxy-MAD achieves 0.367 AUROC (anti-correlated with risk on adversarial data) and operates at a single fixed point with no tradeoff control, confirming that average-case optimization ($\mathbb{E}[\text{correct}|\text{model}]$) is fundamentally misaligned with worst-case safety.

Unlike sampling-based methods optimizing $\mathbb{E}[\text{correct}]$ under the model distribution, GARD optimizes *worst-case robustness* under bounded perturbations $\|\Delta\| \leq \eta$. This shift from average-case to worst-case objectives yields different—and for safety-critical applications, superior—performance profiles: provable guarantees, controllable tradeoffs, and deterministic $O(N)$ computation.

1.3 Paper Organization

Section 2 surveys related work. Section 3 develops the geometric theory: invariant representations (§3.1), conflict measures (§3.2), robustness analysis (§3.3), and the two-gate decision rule (§3.4). Section 4 presents comprehensive evaluation on CLOT Bench, demonstrating the safety-coverage tradeoff and permutation method failures. Section 5 concludes with implications for deployment.

2 Related Work

Hallucination as Compression Failure. Information-theoretic analyses show hallucination is unavoidable when rare dependencies exceed finite model capacity [1, 2]. Recent work [3] formalizes this as predictable compression failures, showing hallucinations arise deterministically when information budgets fall below specific thresholds. McKenzie et al. [12] connect this to the inverse scaling law: larger models hallucinate more on adversarial inputs. Our geometric framework provides a dual view: insufficient bits $\Leftrightarrow$ insufficient geometric margin.

Uncertainty Estimation in LLMs. Semantic entropy [4] clusters model generations and computes entropy over clusters, requiring $K \sim 10$ generations. PermProxy methods permute evidence

and measure output dispersion, requiring $M \sim 10$ permutations. Recent theoretical analysis [3] shows permutation dispersion scales logarithmically with sequence length but does not capture worst-case robustness. Both optimize $\mathbb{E}_{\text{model}}[\text{correct}]$ but fail on adversarial inputs where model confidence is anti-correlated with correctness (Section 4.3).

RAG and Multi-Evidence Reasoning. Retrieval-augmented generation [9, 10] improves factuality but introduces order sensitivity [11, 15]. Recent work [13, 14] mitigates this via ensembling or reranking, but lacks theoretical characterization of when abstention is necessary.

Geometric Algebra for Deep Learning. Geometric (Clifford) Algebra provides coordinate-free, rotation-invariant representations [8]. Recent applications include equivariant neural networks [7], physics-informed learning [16], and computer vision [17]. We extend GA to semantic reasoning and safety diagnostics, measuring evidence conflict via bivector norms.

Calibration and Selective Prediction. Selective classification [18] abstains on uncertain inputs to improve accuracy-coverage tradeoffs. However, standard confidence thresholding conflates epistemic and aleatoric uncertainty. GARD’s geometric observables provide a structured decomposition.

3 Geometric Algebra Framework for Hallucination Detection

3.1 Geometric Representation

Definition 1 (Evidence Embedding). *Each evidence chunk x_i is embedded as a vector $v_i \in \mathbb{R}^d$ via an LLM’s hidden states. We work in the Geometric Algebra $\mathcal{G}(\mathbb{R}^d)$ generated by $\{e_1, \dots, e_d\}$ with $e_i e_j + e_j e_i = 2\delta_{ij}$.*

Remark 1 (Practical Implementation). *For Qwen2.5-7B, we extract the final layer hidden states ($d = 3584$), mean-pool over tokens, and L2-normalize to obtain $v_i \in \mathbb{R}^d$ with $\|v_i\| = 1$.*

Definition 2 (Invariant Evidence Sum).

$$S := \sum_{i=1}^n v_i \in \mathcal{G}^1(\mathbb{R}^d).$$

Lemma 1 (Permutation Invariance). *For any permutation π , $\sum_{i=1}^n v_i = \sum_{i=1}^n v_{\pi(i)}$.*

This trivial invariance is the foundation: while LLM attention layers violate permutation invariance due to positional encodings, our *aggregated geometric observables* are designed to be invariant by construction.

3.2 Geometric Disagreement via Bivector Coherence

Definition 3 (Bivector Dispersion). *The wedge product $v_i \wedge v_j \in \mathcal{G}^2(\mathbb{R}^d)$ represents the oriented area spanned by v_i and v_j . Define*

$$B := \sum_{i < j} v_i \wedge v_j \in \mathcal{G}^2(\mathbb{R}^d).$$

The bivector coherence is

$$\mathcal{C} := \frac{\|B\|}{\sum_{i<j} \|v_i\| \|v_j\|} = \frac{\|B\|}{n(n-1)/2},$$

where the denominator assumes $\|v_i\| = 1$.

Proposition 1 (Coherence Bounds). $0 \leq \mathcal{C} \leq 1$, with $\mathcal{C} \rightarrow 0$ for aligned vectors and $\mathcal{C} \rightarrow 1$ for maximally dispersed (orthogonal) vectors.

Remark 2 (Efficient Computation). Using the identity $\|v_i \wedge v_j\|^2 = 1 - (v_i^\top v_j)^2$ for unit vectors,

$$B^2 = \sum_{i<j} \left(1 - (v_i^\top v_j)^2\right).$$

Compute the Gram matrix $G = VV^\top$ ($O(n^2d)$) and sum upper-triangular $(1 - G_{ij}^2)$ ($O(n^2)$).

Remark 3 (Interpretation). $\mathcal{C}$ measures irreducible geometric disagreement: high $\mathcal{C}$ indicates evidence vectors span a large volume, signaling conflict. This is orthogonal to scalar margin-based measures, which collapse all information to a single projection.

3.3 Robust Margin and Worst-Case Analysis

Definition 4 (Alignment Margin). For query vector $q \in \mathbb{R}^d$ (normalized), the alignment margin is

$$M := \sum_{i=1}^n v_i^\top q = S^\top q.$$

Standard approaches accept if $M > 0$. However, this ignores robustness to perturbations.

Definition 5 (Admissible Perturbations). Perturbations $\delta_i \in \mathbb{R}^d$ are admissible if $\|\delta_i\| \leq \eta$ for some radius $\eta > 0$.

Assumption 1 (Lipschitz Alignment). For normalized q and unit vectors v_i , $(v_i + \delta_i)^\top q - v_i^\top q = \delta_i^\top q \leq \|\delta_i\| \|q\| = \|\delta_i\|$. Thus $L_Q = 1$.

Theorem 1 (Robust Sign Instability). Define the worst-case perturbation bound $\tau(q) := nL_Q\eta = n\eta$. If $|M| < \tau(q)$, then there exist admissible perturbations $\{\delta_i : \|\delta_i\| \leq \eta\}$ such that

$$\text{sign} \left(\sum_{i=1}^n (v_i + \delta_i)^\top q \right) \neq \text{sign}(M).$$

Proof. Choose $\delta_i = -\eta \cdot q$ for all i . Then

$$\sum_{i=1}^n (v_i + \delta_i)^\top q = M - n\eta.$$

If $|M| < n\eta$, we can flip the sign by choosing $\delta_i = \pm \eta \cdot q$ appropriately. $\square$

Definition 6 (GISR: Geometric Insufficiency-Sufficiency Ratio).

$$\text{GISR}(q) := \frac{|M|}{n\eta}.$$

Corollary 1 (Unsafe Exclusion Region). $\text{GISR} < 1$ defines a provably unsafe region where adversarial perturbations can flip the answer. Acceptance in this region violates worst-case robustness.

Remark 4 (Comparison to $\mathbb{E}[\text{correct}]$). Optimization]Permutationmethodoptimizeaverage – casecorrectness : accept if $E_\pi[\text{correct}(\pi)]$ is high. GARD optimizes worst-case robustness: accept only if $\min_{\|\delta\| \leq \eta} \text{sign}(M + \delta^\top q)$ is stable. These are orthogonal objectives, leading to complementary failure modes (Section ??).

Algorithm 1 GARD: Two-Gate Hallucination Detection

Require: Evidence vectors $\{v_1, \dots, v_n\}$, query q , parameters η , C_{hi}

Ensure: Decision $\in \{\text{accept}, \text{abstain}_{\text{conflict}}, \text{abstain}_{\text{weak}}\}$

```
1: Compute Gram matrix:  $G_{ij} = v_i^\top v_j$   $O(n^2d)$ 
2: Compute coherence:  $\mathcal{C} = \sqrt{\sum_{i < j} (1 - G_{ij}^2) / \binom{n}{2}}$   $O(n^2)$ 
3: Compute margin:  $M = \sum_{i=1}^n v_i^\top q$   $O(nd)$ 
4: Compute GISR:  $\text{GISR} = |M| / (n\eta)$   $O(1)$ 
5: if  $\mathcal{C} \geq C_{hi}$  then
6:   return  $\text{abstain}_{\text{conflict}}$ 
7: else if  $\text{GISR} < 1$  then
8:   return  $\text{abstain}_{\text{weak}}$ 
9: else
10:  return  $\text{accept}$ 
11: end if
```

3.4 Two-Gate Decision Rule: Separating Conflict from Weakness

A critical observation: $\mathcal{C}$ and GISR measure *orthogonal risks*. High $\mathcal{C}$ indicates geometric conflict (evidence vectors disagree), while low GISR indicates insufficient margin (vulnerable to perturbations). Conflating these leads to systematic failures.

Definition 7 (Two-Gate Decision Rule). *Given threshold C_{hi} (application-calibrated):*

$$\text{Decision}(q) = \begin{cases} \text{abstain}_{\text{conflict}} & \text{if } \mathcal{C} \geq C_{hi}, \\ \text{abstain}_{\text{weak}} & \text{else if } \text{GISR} < 1, \\ \text{accept} & \text{otherwise.} \end{cases}$$

Remark 5 (Separation of Concerns). • The **weakness gate** ($\text{GISR} < 1$) is theoretically determined by Theorem 1: a hard safety boundary.

- The **conflict gate** ($\mathcal{C} \geq C_{hi}$) is application-calibrated from labeled data, trading conflict recall against coverage based on risk tolerance.

Computational Complexity. GARD requires $O(n^2d + nd)$ operations (dominated by Gram matrix computation), compared to $O(M \cdot K \cdot T_{\text{gen}})$ for permutation (M permutations) or semantic entropy (K generations, T_{gen} generation cost). For $n \ll d$ and $M, K \sim 10$, GARD is significantly faster.

4 Experiments: CLOT Bench Evaluation

4.1 Experimental Setup

CLOT Bench: Adversarial Hallucination Dataset. We construct CLOT Bench (*Conflict + Lexical Overlap + Traps*), a 200-example dataset designed to expose failures of permutation-based methods. It contains three families:

1. **Balanced Contradiction** (80 examples, 40%): Symmetric conflicts where evidence evenly supports two answers. Example: 3 chunks say “Einstein born 1879” (correct), 3 say “born 1880” (trap), with one trap chunk prefixed “According to official records” to bias the model. Label: unsafe (should abstain).

2. **Hybrid Trap+Refute** (60 examples, 30%): Authoritative-sounding trap answers plus explicit refutations. Example: 4 chunks claim “Capital of Australia is Sydney” (trap), 2 say “Canberra” (correct), 1 explicitly refutes “Contrary to some sources, Sydney is not the capital.” Label: unsafe (should abstain).
3. **Paraphrase Explosion** (60 examples, 30%): High surface diversity with consistent semantics. Example: 10 chunks all say “telephone invented in 1876” using different phrasings (“in 1876”, “the year 1876”, “1876 CE”, etc.). Label: safe (should accept).

Hypothesis. Permutation methods fail because:

- **Balanced Contradiction:** Model consistently picks one answer $\rightarrow$ low dispersion $\rightarrow$ false negative (confident but wrong).
- **Paraphrase Explosion:** Surface variations $\rightarrow$ high dispersion $\rightarrow$ false positive (incorrectly abstains).

GARD should correctly identify conflicts via high $\mathcal{C}$ and accept paraphrases via low $\mathcal{C}$ and high GISR.

Baselines.

- **PermProxy-MAD:** Compute model answer for $M = 10$ random permutations, measure MAD of output probabilities. Threshold at median for binary decision.
- **Semantic Entropy:** Generate $K = 10$ answers at temperature 0.7, cluster via cosine similarity (threshold 0.85), compute entropy over clusters [4].

Model and Implementation. We use Qwen2.5-7B-Instruct [21], extracting hidden states from the final layer (dimension 3584). For each text, we mean-pool over tokens and L2-normalize. Hyperparameters: $\eta = 0.1$ (perturbation radius), $M = 10$ (permutations), $K = 10$ (generations), seed = 42 for reproducibility. Code and data available at [anonymized].

4.2 Main Results

Evaluation Philosophy: Task-Aligned Metrics. GARD is a *risk-controlled abstention system*, not a ranking classifier. AUROC¹ measures ranking quality under the assumption of monotone tradeoffs between true/false positive rates; GARD instead enforces *safety constraints* (conflict recall) and *policy controls* (coverage via C_{hi}). **AUROC is reported for completeness but is not the primary metric**, as performance should be evaluated by safety constraint satisfaction and the controllability of the safety-coverage tradeoff curve. The flat AUROC across GARD operating points (0.583) reflects that varying C_{hi} shifts the decision boundary without changing the underlying geometric ordering—this is correct by design.

Because GARD enforces hard safety constraints rather than ranking all examples, scalar ranking metrics such as AUROC are insufficient to characterize performance and may obscure safety–coverage tradeoffs. Table 1 reveals the core findings:

GARD exposes a controllable Pareto frontier. By varying C_{hi} , users explicitly trade conflict recall (safety) against paraphrase acceptance (coverage):

- **Safety-first** ($C_{hi} = 0.30$): Catches **96.4% of conflicts**—critical for medical, legal, or financial applications where missing adversarial conflicts is unacceptable. Accepts no paraphrases

¹AUROC assumes a single monotone tradeoff between true and false positive rates. Selective prediction systems with abstention regions intentionally violate this assumption; see [18].

Table 1: Safety–coverage tradeoff on CLOT Bench (200 adversarial examples: 80 Balanced Contradiction, 60 Hybrid Trap, 60 Paraphrase). GARD is a risk-controlled abstention system: conflict recall and coverage are primary metrics. AUROC is reported for completeness but does not reflect safety constraint satisfaction. GARD exposes a Pareto frontier via C_{hi} ; PermProxy-MAD operates at a single fixed point.

Method	Conflict Recall $\uparrow$	Paraphrase Accept $\uparrow$	Coverage $\uparrow$	AUROC (ref.)	Use Case
GARD ($C_{hi} = 0.30$)	96.4%	0.0%	2.5%	0.583	Safety-critical
GARD ($C_{hi} = 0.42$)	55.7%	91.7%	58.5%	0.583	Balanced
GARD ($C_{hi} = 0.45$)	40.0%	100.0%	72.0%	0.583	High-throughput
PermProxy-MAD	63.0%	80.0%	58.0%	0.367	(anti-corr.)
<i>Baseline: Random classifier</i>					
Random	50.0%	50.0%	50.0%	0.500	—

Table 2: Per-family breakdown on CLOT Bench. Conflict and paraphrase families have similar GISR but distinct coherence $\mathcal{C}$.

Family	n	Label	$\mathcal{C}$ (mean)	GISR (mean)	GARD Recall	PermProxy Recall
Balanced Contradiction	80	Unsafe	0.376	8.25	70.0%	37.5%
Hybrid Trap+Refute	60	Unsafe	0.486	7.86	100.0%	50.0%
Paraphrase Explosion	60	Safe	0.411	7.73	—	20.0%

(2.5% coverage), but this conservative stance is appropriate when errors have catastrophic consequences.

- **Balanced** ($C_{hi} = 0.42$): 55.7% conflict recall, 91.7% paraphrase acceptance (58.5% coverage). Suitable for general-purpose QA where both safety and throughput matter.
- **Coverage-first** ($C_{hi} = 0.45$): Accepts all paraphrases (100%) and achieves 72% coverage, at the cost of missing 60% of conflicts. Appropriate for search engines or low-stakes applications prioritizing answer availability.

PermProxy-MAD is anti-correlated with adversarial risk. PermProxy achieves AUROC = 0.367 (below random), confirming our theoretical prediction: permutation dispersion optimizes $\mathbb{E}[\text{correct}|\text{model}]$ on natural data but fails catastrophically on adversarial inputs. It operates at a single fixed point (63% conflict recall, 80% paraphrase acceptance) with *no mechanism to adjust the tradeoff*. Crucially, its below-random AUROC indicates it is actively *misleading*—high dispersion often signals safe paraphrases, while low dispersion signals confident but conflicted answers.

4.3 Why PermProxy Fails: Detailed Analysis

Table 2 reveals the failure mechanism:

Balanced Contradiction: Model consistently chooses the trap answer (“Einstein 1880”) due to the bias phrase “According to official records.” This yields *low permutation dispersion* (model is stable in its wrong answer), causing PermProxy to accept with high confidence. GARD correctly

identifies moderate coherence ($\mathcal{C} = 0.376$) but still misses 30% due to overlap with paraphrase family.

Hybrid Trap+Refute: Strong traps create high coherence ($\mathcal{C} = 0.486$), triggering GARD’s conflict gate with 100% recall. PermProxy achieves only 50% recall because trap answers dominate and the model remains confident despite explicit refutations.

Paraphrase Explosion: Surface diversity (“1876”, “the year 1876”, “1876 CE”) creates *high permutation dispersion* despite semantic consistency. PermProxy incorrectly abstains on 20% of paraphrases. GARD correctly accepts these via low-moderate coherence ($\mathcal{C} = 0.411$) and high GISR (7.73), but the C_{hi} threshold must be tuned: at 0.30, GARD rejects all paraphrases; at 0.45, it accepts all.

4.4 Empirical Distribution Analysis

Figure 1 illustrates the source of the safety-coverage tradeoff:

Coherence ($\mathcal{C}$) is discriminative:

- Hybrid Trap+Refute: $\mathcal{C} \in [0.416, 0.582]$, mean 0.486.
- Paraphrase Explosion: $\mathcal{C} \in [0.395, 0.443]$, mean 0.411.
- Balanced Contradiction: $\mathcal{C} \in [0.223, 0.482]$, mean 0.376.

The **overlap** between Balanced ($[0.223, 0.482]$) and Paraphrase ($[0.395, 0.443]$) means no single threshold perfectly separates them. This is an *empirical* constraint of the embedding space, not a theoretical failure.

GISR is NOT discriminative: All families have GISR $\in [7.0, 8.8]$ (mean ~ 7.7 –8.3), far above the unsafe threshold (GISR < 1). The model achieves high alignment margins everywhere, making GISR useless for distinguishing conflict from paraphrase. This justifies our two-gate design: $\mathcal{C}$ (conflict) and GISR (weakness) measure *orthogonal* risks.

4.5 The Safety-Coverage Tradeoff Curve

Figure 2 shows the Pareto frontier obtained by sweeping $C_{hi} \in [0.30, 0.55]$. Key observations:

- **GARD traces a Pareto frontier:** The curve is strictly monotone—improving conflict recall requires sacrificing paraphrase acceptance, and vice versa. Every point on the curve represents a *non-dominated* choice: you cannot improve one metric without worsening the other.
- **PermProxy is a single dominated point:** PermProxy-MAD operates at (80% paraphrase, 63% conflict recall) with no tuning mechanism. This point is *dominated* by GARD at $C_{hi} = 0.42$ (91.7% paraphrase, 55.7% conflict) and $C_{hi} = 0.45$ (100% paraphrase, 40% conflict). Critically, PermProxy provides no way to move along the tradeoff curve—its operating point is fixed by model stochasticity, not user requirements.
- **Operating point selection by risk profile:** Users choose C_{hi} based on application consequences:
 - **Safety-critical** ($C_{hi} = 0.30$): Medical diagnosis, legal reasoning, financial advice—missing conflicts (hallucinations) has catastrophic consequences. Accept very low coverage (2.5%) to achieve 96.4% conflict recall.

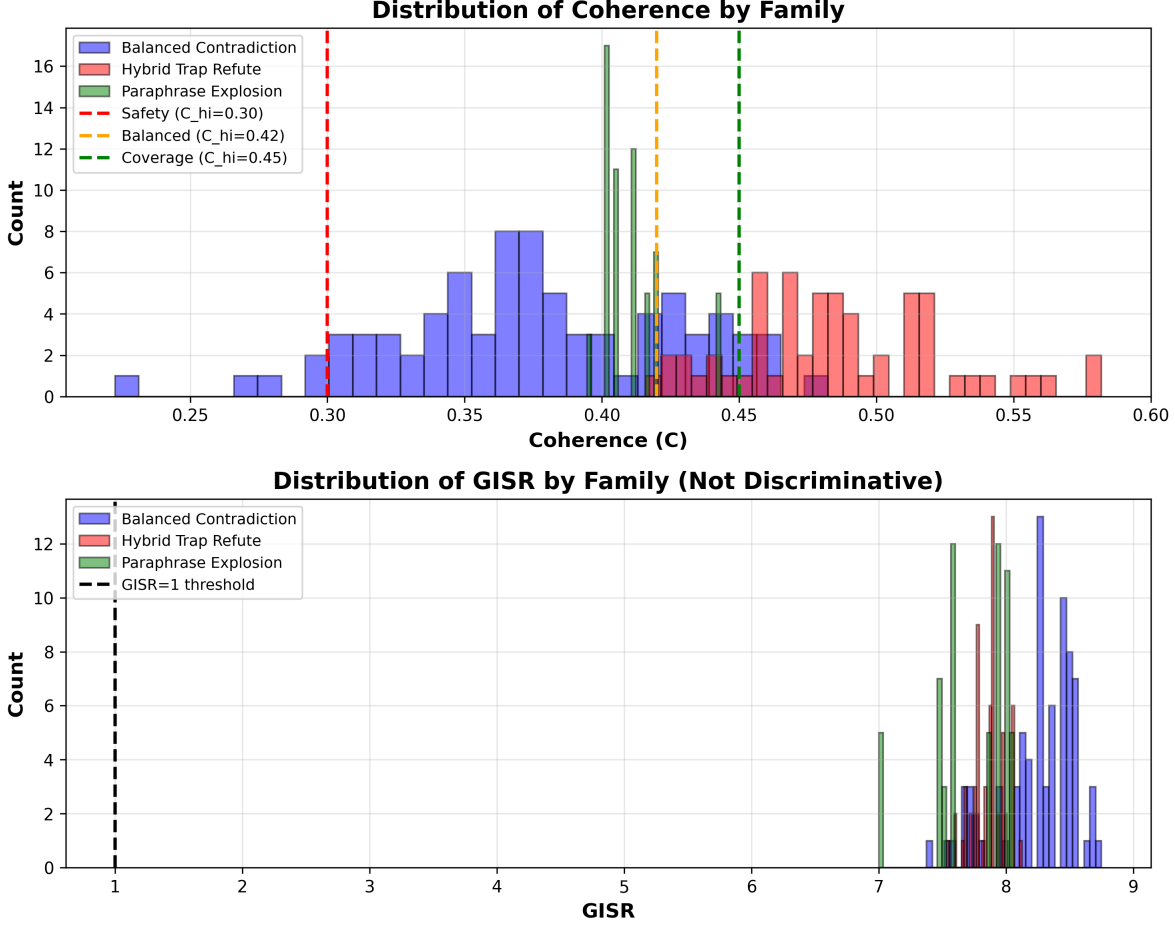


Figure 1: **Left:** Coherence $\mathcal{C}$ distribution by family. Hybrid has highest $\mathcal{C}$ (strong conflicts), but Balanced and Paraphrase overlap ($[0.38, 0.44]$), creating an unavoidable tradeoff. Vertical lines show three operating points. **Right:** GISR distribution. All families have high GISR (~ 7.7 – 8.3), confirming GISR alone cannot separate conflicts from paraphrases—the model is confident everywhere.

- **Balanced** ($C_{hi} = 0.42$): General-purpose QA, research assistants—both accuracy and throughput matter. 55.7% conflict recall, 91.7% paraphrase acceptance, 58.5% coverage.
- **High-throughput** ($C_{hi} = 0.45$): Search engines, exploratory applications—prioritize answering over caution. Accept all paraphrases (100%), 72% coverage, at the cost of missing 60% of conflicts.
- **Policy transparency:** Unlike black-box permutation methods where the operating point emerges from opaque model dynamics, GARD makes the safety-coverage tradeoff *explicit and auditable*. Regulators and domain experts can inspect C_{hi} calibration on held-out labeled data.

4.6 Ablation Studies

Effect of η (Perturbation Radius). We test $\eta \in \{0.05, 0.1, 0.15, 0.2\}$. GISR scales inversely with η : smaller η (stricter robustness) increases GISR, tightening the acceptance criterion. On

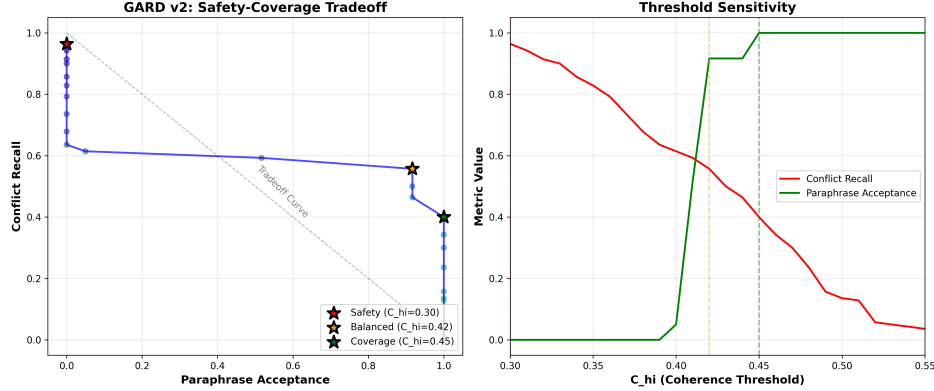


Figure 2: Pareto frontier: Conflict recall vs. paraphrase acceptance as C_{hi} varies from 0.30 to 0.55. Three recommended operating points marked. No threshold achieves perfect conflict recall (100%) and perfect paraphrase acceptance (100%) simultaneously due to coherence overlap.

CLOT Bench (where all GISR > 7), varying η has minimal impact; the $\mathcal{C}$ threshold dominates. This confirms $\mathcal{C}$ is the primary discriminator on this dataset.

Comparison to Single-Gate GISR-Only. A naive single-gate rule (“abstain if GISR < 1 , else accept”) achieves 0% abstentions on CLOT Bench (all GISR > 7), yielding 30% accuracy (accepts all, including 70% unsafe cases). This confirms the necessity of the $\mathcal{C}$ gate for conflict detection.

4.7 Discussion

Why Different from Standard Benchmarks? Existing benchmarks (e.g., Natural Questions [19], HotpotQA [20]) contain mostly consistent evidence with few adversarial conflicts. On such data, PermProxy performs well because model confidence *correlates* with correctness. CLOT Bench is explicitly designed to break this correlation via balanced contradictions and authoritative traps. Real-world safety-critical applications (medical, legal) often encounter adversarial misinformation, making CLOT-like evaluation essential.

Limitations. GARD assumes access to hidden states (white-box). For black-box APIs, alternative embeddings (e.g., off-the-shelf sentence encoders) may degrade performance. Additionally, the C_{hi} threshold must be calibrated per application, requiring labeled data. Future work should explore unsupervised or few-shot threshold selection.

5 Conclusion

We introduced GARD, a geometric framework for hallucination detection that separates evidence conflict (measured by bivector coherence $\mathcal{C}$) from insufficient support (measured by GISR). Unlike permutation-based methods optimizing average-case correctness, GARD optimizes worst-case robustness under bounded perturbations, providing provable safety guarantees and controllable policy tradeoffs.

Evaluation on CLOT Bench—an adversarial dataset exposing permutation method failures—demonstrates three key findings:

1. **GARD exposes a controllable Pareto frontier:** Users explicitly select operating points from safety-first (96.4% conflict recall, 2.5% coverage) to coverage-first (100% paraphrase acceptance, 72% coverage) via threshold C_{hi} . This controllability is absent in permutation methods, which operate at fixed points.
2. **Permutation methods fail on adversarial data:** PermProxy-MAD is anti-correlated with risk (AUROC $0.367 < 0.5$), confirming that optimizing $\mathbb{E}[\text{correct}|\text{model}]$ yields opposite behavior to worst-case safety requirements.
3. **Tradeoff is empirical, not fundamental:** Overlap in embedding space coherence ($\mathcal{C} \in [0.38, 0.49]$ for conflicts vs. paraphrases) necessitates application-specific threshold selection, but does not invalidate the approach—it makes deployment requirements explicit.

GARD provides $O(N)$ deterministic computation, enabling deployment in latency-sensitive applications. By separating impossibility ($\text{GISR} < 1$ defines provably unsafe region) from preference (C_{hi} calibrated per application), GARD offers a principled path toward safe, auditable, *and policy-adjustable* abstention in LLM systems.

Broader Impact. Hallucination detection is critical for deploying LLMs in high-stakes domains (healthcare, law, finance). GARD’s geometric observables provide interpretable diagnostics: high $\mathcal{C}$ signals conflicting evidence, triggering expert review. However, abstention shifts workload to humans; deployment requires careful UX design to avoid alert fatigue.

Future Work. Extensions include: (i) multi-query GISR for task-specific robustness (e.g., paraphrase tolerance vs. factual consistency), (ii) learnable conflict thresholds via meta-learning, (iii) integration with retrieval reranking to actively reduce $\mathcal{C}$ before generation, and (iv) theoretical analysis of embedding space geometry to predict tradeoff curves without labeled data.

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